Counterfactual-Guided Diversified Path Integration for Transferable Neuron Attribution Attacks

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❖ *Contribution I : Counterfactual-guided neuron attribution in the latent space*

(1) We iteratively generate **counterfactual paths** by **minimizing** the discrepancy between the intermediate-layer representations of benign and counterfactual samples.

$$\boldsymbol{\gamma}^{t+1} = \boldsymbol{\gamma}^{t} - \frac{\partial \mathcal{L}_G(\boldsymbol{\gamma}, \boldsymbol{x}^{cf})}{\partial \boldsymbol{\gamma}^{t}} \cdot \frac{\eta}{W_t}$$

(2) Leveraging the counterfactual paths, we integrate along them and **extend counterfactual-guided neuron attribution from the input layer to latent layers.**

$$A_{y_j}^{cf} = \sum_{i=1}^{P} \int_0^1 \frac{\partial F(\boldsymbol{\gamma}^t)}{\partial \boldsymbol{\gamma}_i^t} \cdot \frac{\partial \boldsymbol{\gamma}_i^t}{\partial t} dt$$

$$= \sum_{i=1}^{P} \int_0^1 \frac{\partial F(\boldsymbol{\gamma}^t)}{\partial y_j(\boldsymbol{\gamma}^t)} \cdot \frac{\partial y_j(\boldsymbol{\gamma}^t)}{\partial \gamma_i^t} \cdot \frac{\partial \gamma_i^t}{\partial t} dt$$

(3) We compute counterfactual-guided latent layer neuron attributions via **Riemann sums**.

$$A_{y_j}^{cf} = \sum_{i=1}^{P} \sum_{t=1}^{n} \frac{\partial F(\boldsymbol{\gamma}^t)}{\partial y_j(\boldsymbol{\gamma}^t)} \cdot \frac{\partial y_j(\boldsymbol{\gamma}^t)}{\partial \gamma_i^t} \cdot (\gamma_i^{t+1} - \gamma_i^t).$$

(4) To simplify computation and improve attack efficiency, **we assume that gradients across layers are linearly independent.**

$$A_{y_j}^{cf} = \left(\sum_{t=1}^{n} \frac{\partial F(\boldsymbol{\gamma}^t)}{\partial y_j(\boldsymbol{\gamma}^t)} \right) \cdot \left(\sum_{t=1}^{n} \sum_{i=1}^{P} \frac{\partial y_j(\boldsymbol{\gamma}^t)}{\partial \gamma_i^t} \cdot (\gamma_i^{t+1} - \gamma_i^t) \right)$$

(5) Counterfactual-guided neuron attribution in latent layers ultimately reduces to the product of the **aggregated attribution** IA and the **latent layer representation difference** Δ between benign samples and the baseline .

$$A_{\boldsymbol{y}}^{cf}(\boldsymbol{\gamma}) = \sum_{y_j \in \boldsymbol{y}} A_{y_j}^{cf}(\boldsymbol{\gamma}) = \sum_{y_j \in \boldsymbol{y}} IA_{y_j}^{cf}(\boldsymbol{\gamma}) \cdot \Delta_{y_j}^{cf}(\boldsymbol{\gamma}).$$

❖ *Contribution II: Counterfactual-guided diversified neuron attribution in the latent space*

(1) To diversify counterfactual path generation, we iteratively generate **gradient-ascent counterfactual paths** by **maximizing the intermediate-layer representation discrepancy** between benign and counterfactual samples.

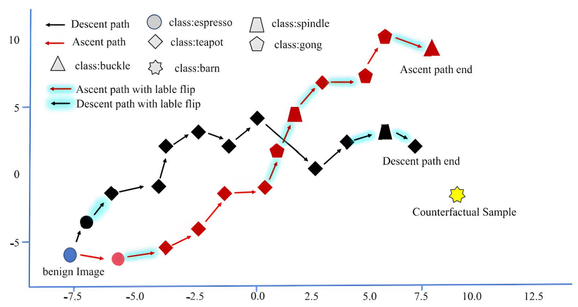
$$\boldsymbol{\gamma}^{t+1} = \boldsymbol{\gamma}^{t} + \frac{\partial \|G(\boldsymbol{\gamma}^t) - G(\boldsymbol{x}^{cf})\|_2^2}{\partial \boldsymbol{\gamma}^{t}} \cdot \frac{\eta}{W_t}.$$

(2) By further **incorporating diverse counterfactual sample reference points**, we derive diverse counterfactual-guided neuron attributions. with C counterfactual sample reference points in total.

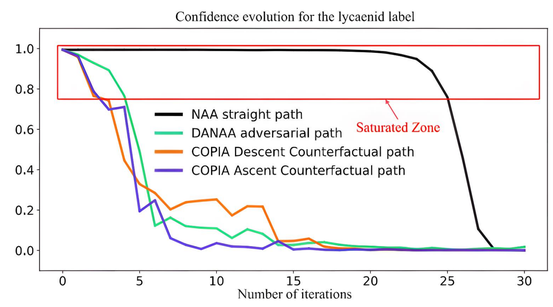
$$\widetilde{A}_{\boldsymbol{y}}^{cf} = \sum_{y_j \in \boldsymbol{y}} \left(\sum_{c=1}^{C} \sum_{\boldsymbol{\gamma} \in \mathbb{P}_c} IA_{y_j}^{cf}(\boldsymbol{\gamma}) \right) \cdot \left(\sum_{c=1}^{C} \sum_{\boldsymbol{\gamma} \in \mathbb{P}_c} \Delta_{y_j}^{cf}(\boldsymbol{\gamma}) \right)$$

$$= \sum_{y_j \in \boldsymbol{y}} \widetilde{IA}_{y_j}^{cf}(\boldsymbol{\gamma}) \cdot \widetilde{\Delta}_{y_j}^{cf}(\boldsymbol{\gamma}).$$

❖ Gradient-ascent and gradient-descent counterfactual paths not only **escape saturated regions faster** (Fig b) but also **traverse distinct decision boundaries** (Fig a), yielding more **precise and robust** neuron attributions.



(a) Gradient-descent vs. gradient-ascent paths.



(b) Different paths for escaping saturation regions.

Contribution

I. We introduce a **counterfactual-guided path into the neuron attribution attack framework**, enabling semantically guided path integration that reduces noise accumulation and improves black-box transferability.

II. We propose a **diversified path integration strategy** that jointly explore different path update directions and an ensemble of counterfactual reference points, leading to more robust and transferable neuron attributions.

III. We develop a **cross-layer attribution aggregation** mechanism that exploits complementary semantic information across multiple layers. Moreover, this general mechanism can be seamlessly integrated into existing single-layer attribution attacks to consistently enhance transferability.

IV. Extensive experiments on the ImageNet dataset demonstrate that **COPIA achieves state-of-the-art performance** in terms of black-box transferability across diverse model architectures, including CNNs, ViTs, and defense models.

Counterfactual path diversification

Cross-layer attribution aggregation(CA)

Improved adversarial transferability

❖ *Contribution IV: Comparison with state-of-the-art (SOTA) methods.*

❖ Surrogate Model: Inc-v3,Inc-v4,IncRes-v2.

■ Attacking Normally Trained and Adversarially Trained Models

Model	Method	Inc-v3	Inv-v4	IncRes-v2	Res-v1$_{50}$	Res-v1$_{152}$	Res-v2$_{50}$	Res-v2$_{152}$	Vgg16	Vgg19	Inv-v3$_{adv}$	InvRes-v2$_{adv}$	Inc-v3$_{ens3}$	Inc-v3$_{ens4}$	IncRes-v2$_{ens3}$	ASR$_{bb}$
Inc-v3	FIA	98.5*	83.1	79.8	72.9	66.4	77.9	73.2	71.6	73.3	53.6	50.6	35.9	36.9	19.1	61.1
	NAA	96.6*	83.3	80.8	71.6	65.1	75.5	75.4	73.0	70.6	56.9	58.1	49.1	50.8	31.4	64.7
	DANAA	97.7*	86.7	84.1	75.2	70.7	79.5	78.6	74.8	72.5	64.4	66.1	54.0	53.7	32.0	68.6
	NEAA	99.1*	87.7	87.4	80.2	73.6	82.7	80.2	78.6	77.3	65.6	64.5	50.9	53.1	32.0	70.3
	SVD	98.6*	81.3	79.9	72.0	64.2	76.9	73.1	71.6	69.9	48.4	52.0	46.2	46.2	30.1	62.4
	MFAA	98.6*	86.5	85.3	79.3	74.7	83.2	80.2	76.8	78.0	60.4	62.9	45.1	44.6	24.7	67.8
	SMP	99.9*	92.1	91.1	80.0	74.5	84.7	82.4	81.4	80.4	58.8	63.4	43.2	41.6	21.7	68.9
	P2FA	**99.9***	**96.7**	**96.5**	**88.8**	83.2	**92.1**	**89.9**	**87.6**	**86.4**	57.0	68.7	33.9	34.9	16.8	71.7
	COPIA	98.8*	94.3	94.2	87.1	**84.8**	89.6	88.1	87.0	85.0	**71.9**	**76.3**	**64.1**	**63.7**	**41.9**	**79.1**
Inc-v4	FIA	84.2	95.7*	78.2	68.9	62.0	75.1	72.2	72.0	69.7	45.2	47.3	38.2	37.4	18.6	59.2
	NAA	81.7	96.0*	75.7	65.3	60.6	72.2	72.3	69.5	66.1	47.3	51.0	47.4	45.2	31.0	60.4
	DANAA	85.2	97.0*	81.8	70.8	65.1	76.9	76.6	73.2	71.2	52.9	59.4	53.2	51.1	31.9	65.3
	NEAA	89.3	98.6*	85.7	79.4	74.2	83.8	82.2	82.0	80.3	64.9	66.5	60.9	58.8	40.0	72.9
	SVD	83.5	96.1*	76.6	72.5	64.0	77.4	72.5	73.5	72.3	47.0	52.7	49.6	48.7	36.1	63.7
	MFAA	82.8	93.3*	78.9	75.0	69.4	79.1	76.5	75.7	73.5	58.4	60.4	54.0	52.5	31.9	66.8
	SMP	92.6	98.8*	88.9	76.2	68.5	84.1	82.2	81.0	79.7	48.8	57.1	46.3	42.8	24.1	67.1
	P2FA	**96.0**	**100.0***	92.3	81.9	72.8	89.0	87.7	82.5	81.9	45.5	54.9	29.9	32.1	14.4	66.2
	COPIA	95.3	99.0*	**93.7**	**87.5**	**83.6**	**89.7**	**89.6**	**88.6**	**87.0**	**71.1**	**76.3**	**70.0**	**66.6**	**51.5**	**80.8**
IncRes-v2	FIA	81.6	78.8	89.6*	71.5	66.5	75.5	71.8	71.8	71.5	60.3	62.9	48.7	44.3	31.7	64.4
	NAA	79.0	75.9	90.6*	68.2	66.0	71.9	70.3	68.7	65.7	56.6	63.6	55.0	52.6	44.4	64.5
	DANAA	82.3	79.6	91.4*	72.9	68.7	76.0	75.4	71.2	70.5	64.2	71.7	63.4	58.8	54.0	69.9
	NEAA	88.6	83.8	95.8*	79.9	76.6	83.0	79.7	79.1	77.9	72.0	75.0	65.8	60.0	52.7	74.9
	SVD	83.1	77.3	94.2*	70.4	65.6	73.9	72.4	70.8	68.0	49.6	58.9	52.0	49.4	41.6	64.1
	MFAA	80.5	79.0	93.9*	73.7	71.0	76.3	74.5	73.0	72.6	64.5	68.7	57.6	54.0	43.6	68.4
	SMP	89.4	87.3	95.8*	82.8	78.4	86.5	84.3	81.0	81.2	70.5	76.8	63.3	56.6	46.4	75.7
	P2FA	93.7	91.3	**99.9***	86.6	80.5	88.4	85.1	84.8	83.8	57.1	69.3	38.6	35.3	21.7	70.5
	COPIA	**96.3**	**94.8**	99.6*	**90.0**	**88.8**	**91.1**	**91.4**	**88.6**	**88.3**	**79.3**	**87.5**	**76.3**	**71.6**	**64.8**	**85.3**

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❖ Surrogate Model: Vit_base,Deit,Visformer.

■ Attacking Vision Transformers Models

Model	Method	Vit$_{base}$	Pit	Cait	Visformer	Deit	Levit$_{256}$	Convit$_{base}$	ASR$_{bb}$
Vit$_{base}$	FIA	93.1	50.8	73.3	54.5	72.1	55.3	74.0	63.3
	NAA	87.7	52.9	71.1	55.5	70.3	57.0	71.0	63.0
	MFAA	97.4	41.3	76.5	48.2	77.2	53.6	77.4	62.4
	P2FA	69.6	46.7	44.5	51.2	43.1	54.2	45.3	47.5
	ATT	99.9	62.9	91.4	68.1	91.8	70.6	92.7	79.6
	FPR	82.3	69.7	65.9	73.1	67.2	77.1	66.3	69.9
	COPIA	**100.0**	**75.7**	**97.3**	**80.1**	**97.2**	**81.3**	**97.0**	**88.1**
Deit	FIA	86.0	75.6	94.7	80.2	99.3	81.9	94.4	85.5
	NAA	84.8	79.0	92.3	80.6	97.7	81.5	91.8	85.0
	MFAA	84.2	65.6	94.6	77.5	99.8	83.3	94.1	83.2
	P2FA	54.5	59.8	60.2	62.1	70.6	65.1	59.5	60.2
	ATT	95.9	88.4	99.2	91.1	100.0	92.4	99.1	94.4
	FPR	82.3	69.7	65.9	73.1	67.2	77.1	66.3	72.4
	COPIA	**97.8**	**88.7**	**99.6**	**92.0**	**100.0**	**94.2**	**99.6**	**95.3**
Visformer	FIA	35.7	45.2	44.8	72.7	41.6	50.7	42.9	43.5
	NAA	40.8	50.7	49.8	78.9	48.4	54.6	47.9	48.7
	MFAA	33.3	57.1	48.4	100.0	43.9	59.6	44.0	47.7
	P2FA	44.5	59.4	52.0	94.8	48.7	63.9	46.8	52.6
	ATT	55.4	82.1	78.0	100.0	75.5	87.3	69.4	74.6
	FPR	47.0	61.6	52.9	91.7	52.2	62.2	52.4	54.7
	COPIA	**65.0**	**83.2**	**80.0**	**98.5**	**78.3**	**87.9**	**74.5**	**78.2**

❖ Generality of Cross-Layer Attribution Aggregation(CA)

■ The **CA** mechanism **improves the transferability** of adversarial examples across **all prior single-layer neuron attribution attacks**.

Model	Method	Inc-v3	Inc-v4	IncRes-v2	Res-v1$_{50}$	Res-v1$_{152}$	Res-v2$_{50}$	Res-v2$_{152}$	Vgg16	Vgg19	Inc-v3$_{adv}$	IncRes-v2$_{adv}$	Inc-v3$_{ens3}$	Inc-v3$_{ens4}$	IncRes-v2$_{ens3}$	ASR$_b$	ASR$_{bb}$
Inc-v3	NAA	96.6*	83.3	80.8	71.6	65.1	75.5	75.4	73.0	70.6	56.9	58.1	49.1	50.8	31.4	67.0	64.7
	NAA$_{CA}$	97.4*	84.8	84.5	76.0	71.4	78.8	77.9	76.3	75.9	59.8	63.5	52.9	52.9	34.6	70.5	68.4
	DANAA	97.7*	86.7	84.1	75.2	70.7	79.5	78.6	74.8	72.5	64.4	66.1	54.0	53.7	32.0	70.7	68.6
	DANAA$_{CA}$	98.5*	91.5	89.8	79.6	74.2	83.5	82.3	77.1	75.0	60.5	64.8	48.0	49.3	27.0	71.5	69.4
	NEAA	99.1*	87.7	87.4	80.2	73.6	82.7	80.2	78.6	77.3	65.6	64.5	50.9	53.1	32.0	72.4	70.3
	NEAA$_{CA}$	99.1*	91.9	92.1	83.7	80.9	85.9	85.4	82.5	81.4	69.1	70.2	57.0	58.3	35.1	76.6	74.9
	S-COPIA	**99.3***	90.5	88.7	81.8	77.9	84.8	82.9	82.7	80.0	69.9	70.8	58.0	57.4	36.3	75.8	74.0
	COPIA	98.8*	**94.3**	**94.2**	**87.1**	**84.8**	**89.6**	**88.1**	**87.0**	**85.0**	**71.9**	**76.3**	**64.1**	**63.7**	**41.9**	**80.5**	**79.1**

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❖ *Contribution III: Cross-Layer Attribution Aggregation(CA)*

To address the lack of diversity in single-layer neuron attribution and **exploit the diversity of multi-layer attributions**, we design a cross-layer attribution aggregation (CA) mechanism. The aggregation from deep to shallow layers proceeds as follows:

(1) For the last layer, the **multi-layer aggregated attribution** $\widetilde{mIA}$ reduces to that of the current layer aggregated attribution $\widetilde{IA}$.

$$\widetilde{mIA}_{\mathbb{L}[L']}^{cf} = \widetilde{IA}_{\mathbb{L}[L']}^{cf}$$

(2) For intermediate layers, the multi-layer aggregated attribution equals the back-propagated attribution flow from the deeper layer plus the current-layer aggregated attribution.

$$\widetilde{mIA}_{\mathbb{L}[l]}^{cf} = \widetilde{IA}_{\mathbb{L}[l]}^{cf} + \mu \cdot \nabla_{F_{\mathbb{L}[l]}} \widetilde{mWA}_{\mathbb{L}[l+1]}^{cf} \Big/ \left\| \nabla_{F_{\mathbb{L}[l]}} \widetilde{mWA}_{\mathbb{L}[l+1]}^{cf} \right\|_1$$

where $\widetilde{mWA}$ denotes the multi-layer weighted neuron attribution importance.

$$\widetilde{mWA}_{\mathbb{L}}^{cf} = \sum_{y_j \in \mathcal{N}_+^{cf}} f_p\left(\widetilde{mA}_{y_j}^{cf}\right) - \lambda \cdot \sum_{y_j \in \mathcal{N}_-^{cf}} f_n\left(\left(-\widetilde{mA}_{y_j}^{cf}\right)\right.$$

❖ Cross-layer attribution aggregation not only **exploits attribution diversity** but also **improves attribution accuracy.**

